MIL-P-43228B
17 November 1977
SUPERSEDING
MIL-P-43228A
21 June 1972

MILITARY SPECIFICATION

PRINTING PRESSES, LETTERPRESS (PLATEN PRESS)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types of electric motor driven printing presses (see 6.1).

1.2 Classification. The presses covered by this specification shall be of the following types and sizes, as specified (see 6.2):

Type I - Platen press, hand fed:
Chase size - 8 x 12 inches (203 by 305 mm).

Type II - Platen press, hand and automatically fed:

Chase size - 10 x 15 inches (254 by 381 mm).
Chase size - 12 x 18 inches (305 by 457 mm).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: US Army Natick Research and Development Command, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3610

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SPECIFICATIONS

FEDERAL

CC-M-1807	-	Motor, Alternating Current, Fractional and Integral Horsepower (500 HP and Smaller).
MMM-A-260	-	Adhesive, Water-Resistant, (For Sealing Waterproofed Paper).
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-1055	-	Barrier Material, Waterproofed, Flexible.

MILITARY

MIL-T-152	-	Treatment, Moisture-and Fungus-Resistant, of Communications, Electronic, and Associated Electrical Equipment.
MIL-C-2212	-	Controllers, Electric Motor, A.C. or D.C. and Associated Switching Devices, Naval Ship Board.
MIL-P-3684	-	Printing, Duplicating, and Book Binding Equipment, Packaging Of.

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-461	-	Electromagnetic Interference Characteristics, Requirement for Equipment.
MIL-STD-462	-	Electromagnetic Interference Characteristics, Measurement Of.

(Copies of specifications, standards, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Standard product. The presses furnished under this specification shall be the manufacturer's current standard products except for those changes required in order to comply with the requirements of this specification. All accessories, components, and tools that are normally offered commercially shall be furnished with each printing press.

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3.2 Materials. Materials not definitely specified shall be of the quality normally used by the manufacturer for printing presses provided the completed items comply with all provisions of this specification (see 6.3).

3.3 Design and construction. The dimensions, weights, and capacities of the presses shall be as specified in tables I and II. The presses shall be either the clamshell or the universal designs and shall be equipped with platen clamps to hold a tympan. The chases shall have a means of being secured to the bed of the press. The press bearers shall be level and parallel to each other. The press frames shall be one piece construction, with provisions for floor mounting. Removable safety guards shall be provided for protection of personnel. The presses shall deliver uniform impressions free from streaking and smearing when tested as specified in 4.3.1 or 4.3.2 as applicable.

TABLE I. Dimensions and weights

	Type I inches (mm) Size 8 x 12 (203 x 305)	Type II inches (mm) Size 10 x 15 (254 x 381)	Type II inches (mm) Size 12 x 18 (305 x 457)
Width, maximum, inches (mm)	42 (1067)	52 (1321)	60 (1524)
Length, maximum, inches (mm)	49 (1245)	69 (1753)	77 (1956)
Weight, maximum, lbs (kg) 1/	1100 (500)	3200 (1450)	4600 (2100)

1/ The weight before crating for shipment.

TABLE II. Capacities

	Type I, inches (mm) Size 8 x 12 (203 x 305)	Type II, inches (mm) Size 10 x 15 (254 x 381)	Type II, inches (mm) Size 12 x 18 (305 x 457)
Paper size	2 x 3 (51 x 76) to 8 x 12 (203 x 305)	2 x 4 (51 x 102) to 10 x 15 (254 x 381)	3-1/2 x 4 (89 x 102) to 13-3/8 x 18-1/8 (340 x 460)

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3.3.1 Type I. The type I press shall be electric motor driven, and hand fed. The platen shall be provided with movable sheet grippers, tympan clamps, and platen impression adjustment screws. The inking unit shall consist of an ink fountain, a solid rotating ink dish, and a minimum of three form rollers. Each form roller shall have two bearing trucks. The inking unit shall be capable of providing uniform distribution over the type form, when tested as specified in 4.4.1. The press shall be equipped with three chases, positive acting brake, and an impression throw-off lever. The press shall have a variable speed drive with a switch for control of the press drive motor up to a speed of 1800 impressions per hour. The press shall be provided with a five-digit impression counter. The allowable impression spoilage shall be a maximum of 8 percent, when tested as specified in 4.4.1.

3.3.2 Type II. The type II presses shall be electric motor driven and capable of both hand and automatic paper feed. The air pump shall be provided with an intake filter. The platen shall be provided with movable sheet grippers, tympan clamps, and impression adjustment screws. The inking unit shall consist of an ink fountain, two vibrating distributor rollers, a solid rotating disc, and a minimum of four form rollers. Each form roller shall have two bearing trucks. The inking unit shall have an "on" "off" ink control. The ink fountain rollers shall be adjustable for light and heavy ink distribution. The presses shall be equipped with three chases, a positive action foot pedal control brake, an impression throw-off lever, and a five-digit impression counter. The presses shall have a variable speed drive. The size 10 by 15 press shall be capable of up to 4000 uniform impressions per hour, and the size 12 by 18 press shall be capable of up to 3500 uniform impressions per hour. The allowable impression spoilage shall be a maximum of 5 percent, when tested as specified in 4.3.2. The inking unit shall provide uniform ink distribution over the type form when tested as specified in 4.3.2. The press shall be equipped with a device to automatically throw the press "off-impression" when a sheet is either improperly fed, improperly delivered, or when the paper supply is exhausted, when tested as specified in 4.3.2.

3.3.3 Controls. All controls shall be mounted in such a manner that they can be operated while the press is running. When specified (see 6.2), motor controls for shipboard use shall conform to MIL-C-2212.

3.3.4 Electrical requirements. All presses shall operate on 115 volt, 60 hertz (Hz), single phase, or 220 volt, 60 Hz, three phase, alternating current, or 440 volt, Hz, three phase, alternating current, as specified (see 6.2).

3.3.4.1 Motors.

3.3.4.1.1 Type I press motors. The motor for type I press shall be continuous duty, minimum 1/4 horsepower (190 W) and conform to CC-M-1807.

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3.3.4.1.2 Type II press motors. The motor for type II press shall be continuous duty, minimum 1-1/2 horsepower (1120 W) and conform to CC-M-1807. The double rotary pump motor shall be a continuous duty, 3/4 minimum horsepower (560 W) and conform to CC-M-1807.

3.3.4.2 Electromagnetic compatibility. When specified (see 6.2), printing presses shall be designed and equipped for electromagnetic compatibility in accordance with class IIB of MIL-STD-461 (see 4.3.3).

3.3.4.3 Moisture and fungus resistant treatment. When specified (see 6.2), electrical components of equipment covered by this specification shall be moisture and fungus resistant treated with material conforming to and applied as specified in MIL-T-152.

3.4 Finish. The presses shall be finished in accordance with the manufacturer's standard commercial practice.

3.5 Marking for identification. The presses shall be marked for identification in accordance with the manufacturer's standard commercial practice and shall include manufacturer's name, date of manufacture, and model number.

3.6 Instruction plates. Instruction plates shall be provided in accordance with the manufacturer's standard commercial practice and shall include warning and caution plates.

3.7 Contract data requirements. The contract data requirements shall be as specified (see 6.2 and 6.2.1).

3.8 Workmanship. Printing presses shall be free from defects such as fractures, splits, punctures, tears, dents, creases, deteriorations, or malformation. The press bearers shall be smooth and free of burrs, sharp edges, or other imperfections.

3.8.1 Welding. The surface of parts to be welded shall be free from oxide, scale, paint, grease, and other foreign matter. Welds shall be continuous (on structural supports), sound, smooth, and free from porosity, cracks, incomplete fusion, and deformation of material. All scale and flux shall be removed from the finished welds.

3.8.2 Threaded fasteners. Threaded fasteners shall not be broken, stripped, fractured, or loose.

3.8.3 Electrical wiring. Wiring shall not be cut, abraded, or have excessive insulation stripped, and shall be properly and tightly joined at terminals.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection. Except as otherwise specified herein, sampling shall be performed in accordance with MIL-STD-105.

4.2.1 Component and material inspection. In accordance with 4.1, the components and materials shall be inspected in accordance with all the requirements of the referenced specifications and standards, unless otherwise excluded, amended, modified, or qualified in this specification.

4.2.2 In-process inspection. Examination shall be made during an appropriate phase of the manufacturing process to determine whether the fungus and moisture resistance protection specified in 3.3.4.3, is completely applied. Whenever nonconformance is noted, correction shall be made to the press and the process.

4.2.3 End item inspection. The sample unit for this inspection shall be one completely fabricated printing press. The press shall include tools and specified optional equipment. Whenever a defect is noted, correction shall be made to the press and the process.

4.2.3.1 Visual examination. Visual examination of the completely fabricated printing press shall be made for defects in table III.

TABLE III. Visual examination for defects

Examine	Defect
Finish	Not finished in accordance with manufacturer's standard commercial practice
Type of press	Not as specified
Platen	Out of alignment, bent, warped Impression surface not smooth

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TABLE III. Visual examination for defects (cont'd)

Examine	Defect
Platen (cont'd)	
Impression adjustment screws	Missing or loose
Sheet grippers	Missing, not removable, or bent
Tympan clamps	Missing, bent, or rough edges
Chase	Missing, bent, or warped
Ink fountain	Missing, broken, bent, or chipped
Vibrating distributor and form rollers	Missing, chipped, cracked, or out of alignment
Bearers	Not level or parallel, or not smooth and free of burrs, sharp edges, or other imperfections
Ink disc	Missing, bent, warped, or pitted
Press brake	Missing, not operable, or not as specified
Speed control	Missing or not as specified
Impression counter	Missing, broken, or not as specified
Impression trip lever	Missing or binds
Rotary pump (type II)	Pump not as specified, out of alignment with motor, or control missing
Wiring	Cut abraded, not completely secured to terminals, or excessive stripping of insulation
Electrical components	Fungus and moisture resistance missing, when specified Motors not as specified, out of alignment
Machined, cast, and sheet metal components	Rough or sharp edges
Welding	Missing, incomplete, burn holes, cracked, fractured, or not fused Slag inclusion, undercut, not smooth and uniform, scale, or flux deposits

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TABLE III. Visual examination of defects (cont'd)

Examine	Defect
Threaded fasteners	Broken, stripped, fractured, or loose
Marking	Missing, incomplete, or illegible
Instruction plates	Missing
Tools	Not available when normally supplied by manufacturer

4.2.3.2 Dimensional and weight examination. Each completely assembled press shall be examined for defects in dimensions and weight. Any deviation from specified requirements shall be classified as a defect and correction shall be made.

4.2.4 End item testing. Each press shall be tested in accordance with 4.3.1 and 4.3.2, as applicable, and 4.3.3, when applicable. When a defect is noted, it shall be corrected and the press shall be completely retested. If the defect cannot be corrected, the press shall be rejected.

4.2.5 Packaging inspection. An examination shall be made of each press to determine that all preservation-packaging, packing and marking comply with section 5 of this specification.

4.3 Tests.

4.3.1 Operation, type I. The press shall be operational tested in two phases. In the first phase the press shall be set up (in accordance with the manufacturer's operating instructions) in a room at 68 to 75°F (20 to 24°C) and 45-50 percent relative humidity. A type form shall be made up in the chase. The form shall consist of five characters: one positioned in each extreme corner of the chase, and the fifth positioned in the center of the chase. The five characters shall be 60-point (21 mm) size, Roman serif capital (M's). The type form shall be imposed and locked in the chase. A tympan shall be clamped upon the platen. The press shall be inked (as specified in the manufacturer's instructions), and operated (off impression) until the ink has been uniformly distributed over the face of the ink disc. The chase shall be locked in the press. The press shall again be operated (off impression) until the ink is uniformly distributed over the type form. An impression shall be made on the manila draw sheet of the tympan. The sheet shall be examined for uniformity of impression and, if necessary, the platen shall be adjusted accordingly. Remove and replace the sheet, and make a new impression. Repeat this procedure until the impression on the sheet is uniform, to determine conformance with 3.3 and 3.3.1. The testing time required for the first phase shall not exceed 15 minutes. The second phase shall be a continuous operation run, at maximum speed, for 1 hour. The type form for this phase shall

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consist of 14 lbs (6.4 kg), foundry or machine-cast characters, 21 picas (89 mm) wide by 30 picas (127 mm) deep, set solid. The paper sheet size shall be 5 inches (127 mm) by 8 inches (203 mm), and the paper shall be 20 lb. (9.1 kg) sulfite bond cut with the grain short. There shall be 2000 sheets placed on the feedboard in lifts of 500 sheets each. The tympan packing shall be as specified in the manufacturer's instructions. The press shall be inked for operation as specified in the manufacturer's instructions. After the press has been adjusted, it shall be operated at maximum speed for 1 hour. The printed sheets shall be examined and counted to determine conformance with 3.3.1.

4.3.2 Operation, type II. The type II presses shall be operational tested in three phases. The first phase shall be the same as the test for the type I press, specified in 4.3.1, to determine conformance with 3.3. The second phase shall also be the same as the test for the type I press, except the paper sheet size shall be 8-1/2 inches (216 mm) by 11 inches (280 mm), the paper quantity shall be 4400 sheets for the size 10 by 15, and 4000 sheets for the size 12 by 18 and the press shall be run on automatic feed and the hopper loaded as necessary to comply with the foregoing requirement. The printed sheets shall be examined and counted to determine conformance with 3.3.2. The third phase shall consist of feeding a sheet of paper positioned so that it will operate the automatic impression (throw-off device) to determine compliance with 3.3.2. The press shall then be operated (on-impression) until the paper supply is exhausted and the press is automatically thrown (off-impression) to determine compliance with 3.3.2.

4.3.3 Electromagnetic compatibility. When electromagnetic compatibility is required, the initial unit of production shall be tested by the contractor in accordance with the test methods CE03 and RE02 of MIL-STD-462. The Government reserves the right to witness tests performed by the contractor or an independent testing agency. The contractor shall furnish the contracting officer written certification that the Interference Control Plan, EMI/EMC Test Plan, the Electromagnetic Test Report, and the requirements meet MIL-STD-461.

5. PACKAGING

5.1 Preservation-packaging, packing, and marking. Preservation-packaging, packing, and marking for level A, B, or C shall be in accordance with MIL-P-3684 (see 6.2), except that for level A pack the contents of each nailed wood shipping container conforming to PPP-B-621 shall be covered with a full-height shroud made of waterproof barrier material conforming to class E-2, H-1, H-5, or M-1, of PPP-B-1055. All seams of the shroud shall be sealed with a 3/4-inch (19 mm) minimum width continuous seam of water-resistant adhesive conforming to MMM-A-260.

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6. NOTES

6.1 Intended use. The type I press is intended for use in production of cards, forms, and letterheads in large quantities. The type II presses are intended for instructional purposes, and use in the automatic production of cards, forms, letterheads, billheads, etc. in large quantities.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type and chase size required (see 1.2).
(c) When motor control is required for shipboard use (see 3.3.3).
(d) Required electrical voltage (see 3.3.4).
(e) When electromagnetic compatibility is required (see 3.3.4.2).
(f) When moisture and fungus resistant treatment is required (see 3.3.4.3).
(g) Contract data requirements (see 3.7 and 6.2.1).
(h) Selection of applicable levels of preservation and packaging and packing (see 5.1).

6.2.1 Contract data requirements. Any data items to be delivered under any contract covered by this specification shall be specifically called for in the contract in accordance with the applicable regulation of the procurement agency (see 3.7 and 6.2).

6.3 Recycled material. It is encouraged that recycled material be used when practicable as long as it meets the requirements of the specification (see 3.2).

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:

Army - GL
Navy - SA
Air Force - 99

Review activities:

Army - MD
DLA - GS

Preparing activity:

Army - GL

Project No. 3610-0253

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE
MIL-P-43228B - Printing Presses, Letterpress (Platen Press)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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